Application of psychological counseling system based on virtual reality technology in college students' psychological counseling

Dongxing Yu[1]*, Qing Ji[2]
[1]School of Education, Sanda University, Shanghai, China.
[2]Counseling and Psychological Services of ECNU, East China Normal University, Shanghai, China.
*Email: yudongxing@sandau.edu.cn

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Abstract*—In recent years, with the continuous acceleration of the social rhythm, from the primary school students who learn to speak in the classroom to the middle school students facing the college entrance examination and the college entrance examination, and even adults facing the pressure of life, there are different degrees of psychological pressure and people's psychological pressure. If these pressures do not know how to release and vent properly, they may develop into serious mental illnesses, which may lead to self-harm or even suicide, thus seriously endangering people's normal lives. This paper aims to study the psychological counseling assistance system based on virtual reality technology. This paper makes a detailed investigation and analysis of traditional psychological counseling methods, understands the shortcomings and deficiencies of traditional psychological counseling methods, and finds out the corresponding solutions, which lays the foundation for the development of a psychological counseling assistance system based on virtual reality technology. Then it systematically analyzes the psychological counseling assistance system of virtual reality technology from three aspects: function, data analysis and data practice. This article focuses on four aspects: user, system function, technology and hardware peripherals required by the system. A comprehensive needs analysis was carried out. The experimental research shows that in the data practice of the psychological counseling assistance system based on virtual reality technology, the case summary of a university for four years was analyzed, and the problems and proportions faced by college students at that time were obtained from it, and it was extended to the participation work. middle. After the crowd, find out their psychological problems and make a summary.

Keywords:Psychological Counseling, Virtual Reality, Auxiliary System, Psychological Safety

I. Introduction

The number of domestic psychological mentors and psychiatrists is inherently scarce. In addition, the country has always been affected by traditional culture, and there are various misunderstandings about psychological counseling. It is a shame to see a psychologist, and I would rather endure it for a long time. Mental suffering and distress, do not go to the psychologist to tell the distress and troubles in the heart, so many people can not solve their psychological problems, or choose no good way, and many people with psychological problems dare not agree with the psychologist. The exchanges with Yi have caused many people's psychological problems to be unresolved, which has seriously affected their own progress.

This paper combines virtual reality technology with psychological counseling. In the research based on psychological counseling assistant system, many scholars have studied it and achieved good results. For example, Wang Shengyuan proposed that virtual reality technology can be combined with a variety of psychotherapy methods to achieve relatively good results in solving some psychological problems. With the vigorous development of virtual reality technology in China in recent years, many researchers spend information on the treatment of mental diseases such as psychophobia [1]. Some scholars analyzed the requirements of virtual reality technology for the production of psychological counseling scenes and put forward the design ideas of psychological counseling and psychological assistance, and concluded that the use of virtual reality technology in the training of psychological counselors is feasible [2]. This paper determines the relevant research on the psychological counseling assistant system based on virtual reality technology, and on this basis, develops a new platform system for psychological counseling and quality.

This paper determines the research direction and data analysis of psychological counseling assistant system based on virtual reality technology. Based on the summary and analysis of a four-year case of a university, this paper obtains the categories of all problems faced by the students of the university, analyzes the proportion, and then extends it to the people after taking part in work. Then obtain and analyze the problems faced by the population, and finally design a psychological counseling system suitable for the use of the population [3].

II. Overview and Application Technology of Virtual Reality

A. Virtual Reality

Virtual reality is a rapidly emerging discipline. The idea of Virtual Reality was first put forward by Jaron Lanier, and was put forward in the paper Ultimate Display in 1965. This article first put forward the three-dimensional display Based on the addition of interactive images and sounds, this is the predecessor of VR technology, which was later promoted and applied by NASA, which accelerated the progress of virtual reality technology. At present, this technology is in a blowout development [4].

Virtual reality technology, our country calls it (Virtual Reality) VR technology for short. As an emerging design and development technology, it combines the advantages of computer graphics technology, simulation technology, artificial intelligence technology, sensors, display and parallel network processing technology [5].

Virtual reality technology is divided into narrow and broad concepts. In the narrow sense, it refers to intelligent or advanced human-machine interface. It uses several senses to conduct real-time simulation and real-time interaction, allowing people to pass through the eyes and ears. Nose to

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feel and recognize computer-simulated touch, smell, and images. The created virtual world promotes human-computer interaction and improves the effectiveness of human-computer interaction. In a broad sense, maturity is the realization of virtual imagination or the simulation of the real world. It simulates the electronic part of the objective world and receives or responds to various organ stimuli in the simulated environment in a natural way. The communication between people and things makes users feel immersive [6].

VR technology has three main research directions: one is to provide an entrance to browse virtual scenes, the other is 3D visual presentation based on computer graphics, and the third is to help and support other fields [7].

B. Characteristics of Virtual Reality Technology

The biggest feature of virtual reality technology is that it allows users to interact with the virtual environment in the most natural way. On the other hand, it is also a design tool for designing another information space. Since virtual reality technology is the interaction between the user and the object in the 3D space created by the computer, in addition to projection, the object can also move freely in the space according to the user's wishes, thereby producing significant meaning. Training and application in the medical industry, construction industry, science and technology, military aerospace, education and teaching [8-9].

Immersion: All the user's actions in the real scene will be fully mapped to the virtual world, including body movements, language, etc. This is also the focus of virtual reality technology research [10].

Interactivity: Interactivity requires that the user's head movement in the real world is in real time to the scene, and their feedback will cause the user's perspective in the virtual scene to change [11].

Imagination:When the user uses the virtual reality system to operate in the real world, in the virtual world, the character has only a stone in his hand. Right now, this stone is going to be thrown out, and at the same time it has to move with the trajectory like the real world [12].

C. Principles of Virtual Reality

The realization goal of virtual reality technology is to use computer-related technology to construct a virtual world similar to reality, which is used to provide customers with a feeling of being in reality and immersing users in it. The principle of the process is as follows:

Definition $S(X)$As a transformation function, data collection for user operation

$$D_0 = S(0) \tag{1}$$

Where 0 is a collection of user operations, D_0 For the transmission of D_0, a function$T_0(X)$ is defined to deform it:

$$D_1 = T_0(D_0) \tag{2}$$

D_1 For the changed operation data, then the data is transmitted to the upper computer for analysis and the function $T_1(x)$ is defined to deform it:

$$D_2 = T_1(D_1) \tag{3}$$

Then use the parsed data to construct the scene and define R_x as a function:

$$V = R(T_1(T_0(S(O)))) \tag{4}$$

By analogy, the data is fed back to the customer in the form of 3D images to achieve the effect of human-computer interaction.

D. Analysis of the Main Functions of the Psychological Counseling Auxiliary System

- User management function

Mainly realize the user's information management, simple delete, change, and assign authority to the user's registered account.

- Query function

The purpose is to realize the query function of the system, and can also realize the query of the past classic cases, and the query of the user's psychological consultation records, etc., in the long-term psychological tracking and psychological analysis of the users, and timely discover the psychological problems of each user and provide timely guidance and treatment to protect the psychological safety of users.

- Psychological test function

It is mainly to provide users with psychological tests. Through simple psychological tests, users can have an evaluation of their own psychology so that users can understand their own psychological conditions in time, find out existing psychological problems as soon as possible and solve them as soon as possible.

- Appointment function

Due to the domestic environmental impact, the number of mental health specialists and psychologists in our country is not large, and with the increasing pressure in various aspects of society and life, there will be more and more users with psychological problems. The number of existing psychologists cannot meet the demand for the number of users. For the sake of efficiency, an appointment method is opened to give full play to the role of psychologists.

- One-to-one psychological consultation function

After completing the appointment and arriving at the appointed time, you can enter the corresponding consultation room, and this completion method is mainly completed by online communication and interaction between the user and the doctor.

- Online classroom function

To realize the psychological popularization of users and the learning of mental health knowledge, in the psychological classroom, the psychological development of the staff and users is mainly accomplished through lectures, videos and text stories.

III. Component of Psychological Counseling Assistance System Based on Virtual Reality Technology

A. Extraction of Environmental Factors

- Input equipment

The simplest and most direct way for users to observe the world with their eyes is to use their hands. Therefore, they need a virtual reality glove, 3D mouse, 3D joystick and other equipment to complete the psychological counseling system.

- Motion and expression capture input device

The purpose of virtual reality technology is to hope that users can be immersed in the virtual scene of the right component, and the psychological counseling assistant Zitong also hopes that users can openly communicate with psychologists. In the process of communication, users will

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consciously and unconsciously make a change of expression and actions, because these actions and expressions are an important way for psychologists to obtain information about Honghu, so it is necessary to recognize and capture the user's expressions and actions. At present, these functions of virtual reality technology are mainly concentrated in virtual reality helmets and glasses. Motion capture can be solved by using cameras and sensors to restore all the user's performance most accurately.

- Voice control equipment

As a way of daily communication, words are already an indispensable part of our lives. Combining virtual reality technology with intelligent voice technology is also a good research direction of virtual reality technology. In real life, we also use voice input devices to perform certain system functions. In the virtual reality scene, it is also hoped that the user can perform similar functions in the virtual scene through voice input. Through such a device, the user can interact with some virtual characters in the virtual scene, thereby enhancing the user's sense of immersion. The difficulty of voice-activated input devices lies in voice recognition. When speech recognition technology reaches a certain level, it will have a huge boost to virtual reality technology.

B. Information Transmission Method

The input device part of the psychological consultation system is separated from the scene component, but the scene component needs to obtain the data collected by the device, so it is in a very important position in the psychological consultation system based on virtual reality technology, and the wired transmission method is too dependent on long data transmission via cable will lead to failure due to the distance being too far during the consultation process.

There are currently three wireless transmission methods: Bluetooth, WIFI, and Zig Bee.

- Bluetooth

As a wireless technology, data can be exchanged between personal domain networks, fixed devices and mobile devices. The earliest Bluetooth technology was released by Ericsson in 1994. At that time, it was only an alternative to the RS232 data line. However, with the progress of the times, Bluetooth technology has also made significant progress. Due to its close range and low range, the speed, low cost and other characteristics make it widely used by many companies, including mobile devices, GPS devices and computer equipment, which are equipped with Bluetooth. Its range is about 10 meters. The current rate is about 25Mb/s.

- Zig Bee technology

It is a local area network protocol based on the IEEE802.15.4 standard. It has the same characteristics as Bluetooth, short distance, low speed, low power consumption and other characteristics including low complexity. But it is more reliable than Bluetooth, because it uses the CSMA/CA mechanism and the CRC check mechanism in the MAC layer. It is mainly used in the fields of automatic control and remote control. Its range is about 10 to About 75 meters, this is related to its power consumption. The current rate only supports a rate of about 250kb/s in the 2.4GHz-based physical layer.

- WIFI technology

WIFI technology is a technology that can connect electronic devices to the same wireless local area network (WLAN), and it is also a short-distance wireless transmission technology. But compared with the other two technologies, its working range is larger than that of Zig Bee and Bluetooth, which can reach about 100 meters. The current rate can reach 54Mbp/s, which is more suitable for daily use.

IV. Practice of Psychological Assistant System Based on Virtual Reality

3DS MAX software provides visual and systematic three-dimensional modeling capabilities, while VRML is a three-dimensional language based on virtual reality technology. It is also used to construct virtual scenes and can achieve virtual scenes and user interaction effects. These two technologies of combination can realize the experiment of psychological counseling assistance system based on virtual reality technology.

A. Consultation Data Experiment

As shown in Table 1, the psychological counseling room of a certain university analyzed the psychological counseling files of the four years from 2017 to 2020, and obtained statistics on the number of college students, gender, and other data. The results showed that the proportion of freshmen and seniors ratio is the largest, and the ratio of freshman girls to boys has reached about 31%, which is very high.

TABLE I. Analysis of the number of boys and girls

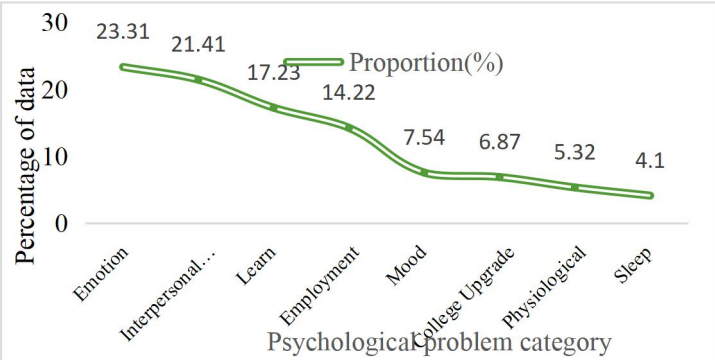
	College freshman	Sophomore	third year in university	Fourth year of university
Number of girls	391	313	224	321
Proportion	31.3%	25.0%	17.9%	25.8%
Number of boys	151	122	95	115
Proportion	31.2%	25.2%	19.6%	23.8%

As shown in Table 1, the number of males taking part in the psychological test in a university in the past 4 years is obviously less than that of females. The reason is that females are delicate and easy to be sentimental, while males can play games and exercise when they encounter troubles. According to the data, most students encounter psychological problems in their freshman and senior year.

B. Case Analysis of Psychological Counseling

As shown in Figure 1, the data of the psychological counseling room of a university during the four years from 2017 to 2020 is classified and summarized. After analyzing and sorting out, the main psychological problems faced by college students at that time were obtained and classified.

Fig. 1. Psychological problem classification diagram



As shown in Figure 1, in the psychological data summary and analysis of a university during the four years from 2017 to 2020, the most problematic is that the emotional problem reached 23.1%, and the least problem is the sleep problem.

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There are a lot of people who are in love at this university and don't know how to deal with social relationships.

V. CONCLUSIONS

After unremitting efforts, this article uses the data obtained from the data practice to complete a complete psychological counseling assistance system based on virtual reality technology. It makes full use of the case in the psychological counseling room of a university and reflects the college students' face at that time. The correct series of problems are reflected in the completed system and these problems are solved. These data are substituted into the system to complete the user's management function, query function, psychological test function, appointment function, one-to-one psychological consultation function and many functions such as online classroom function.

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REFERENCES

[1] Wang Shengyuan. New technology of psychological consultation and treatment: virtual reality technology[J]. *Science Education Journal*, 2019, 000(002):165-166.

[2] Hu Dongtao, Huang Lang, Zhou Huanming, et al. Design and implementation of training system for non-coal mine rescue team based on virtual reality technology[J]. *China Work Safety Science and Technology*, 2017, 013(002):171-175 .

[3] Du Yuan, Xie Xuchen, Guo Lei, etc. Distributed Data Quick Search System Based on Augmented Reality Technology[J]. *Electric Power System Equipment*, 2019, 000(010):151 -152.

[4] LIU Hui, ZHANG Nan. The Construction of Urban Architectural Narrative Space Based on Virtual Reality Technology%[J]. *Journal of Hunan Institute of Science and Technology (Natural Science Edition)*, 2019, 032 (002): 67-72.

[5] Xu Yuren (Yu-Jen Hsu), Yu-Yun Wen. A Case Study on How a User's High Altitude Experience in a Virtual Reality Game Affects Emotional Experiences[J]. *Journal of Digital Learning Technology*, 2017, 9(2): 85-108.

[6] Serra-Pla J F , Pozuelo M , Richarte V , et al. Treatment of attention deficit hyperactivity disorder in adults using virtual reality through a mindfulness programme[J]. *Revista de Neurologia*, 2017, 64(s01):117-122.

[7] Roberts, A. C. , You, W. Y. , Seah, H. S. , Chan, E. , & Christopoulos, G. I. . (2019). Assessing the suitability of virtual reality for psychological testing. Psychological Assessment, 31(3), 318-328.

[8] Riva, G. Letter to the Editor: Virtual reality in the treatment of eating and weight disorders[J]. *Psychological Medicine*, 2017:2567-2568.

[9] Liu Min. Research on Intelligent Gesture Recognition Based on Neural Network and Virtual Reality Technology% Research on Intelligent Gesture Recognition Based on Neural Network and Virtual Reality Technology[J]. *Journal of Changchun Normal University (Natural Science Edition)*, 2018, 037(003): 69-74.

[10] Wang Hao, Zeng Hong, Ni Wenli, etc. Marine auxiliary disassembling training system based on virtual reality[J]. *Journal of Jiangsu University of Science and Technology (Natural Science Edition)*, 2017 , 031(001):44-48.

[11] Li Qian, Tian Jiya. 3D laser scanning image reconstruction based on virtual reality technology[J]. *Laser Magazine*, 2017, 038(008):139-142.

[12] Al-Shammari M , Gao T H , Mohammed R K , et al. Attention enhancement system for college students with brain biofeedback signals based on virtual reality[J]. *Multimedia Tools and Applications*, 2021:1-16.

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